

## Proofs for SLR Algebraic Equivalencies

$$y_i = b_0 + b_1 x_i + e_i$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$\hat{y}_i = b_0 + b_1 x_i$$

$$\text{Prove: } \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

$$\text{Start by expanding the square: } (x_i - \bar{x})^2 = x_i^2 - 2x_i\bar{x} + \bar{x}^2$$

$$\text{Summing over } i: \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2$$

$$\text{Since } \bar{x} \text{ is constant, } \sum_{i=1}^n \bar{x}^2 = n\bar{x}^2 \text{ and } \sum_{i=1}^n x_i = n\bar{x}$$

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - 2\bar{x}(n\bar{x}) + n\bar{x}^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

$$\boxed{\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2}$$

$$\text{Prove: } \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$$

$$\text{Expand the product: } (x_i - \bar{x})(y_i - \bar{y}) = x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}$$

$$\text{Sum over } i: \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - \bar{y} \sum x_i - \bar{x} \sum y_i + \sum \bar{x} \bar{y}$$

$$\text{Use: } \sum x_i = n\bar{x}, \sum y_i = n\bar{y}, \sum \bar{x} \bar{y} = n\bar{x} \bar{y}$$

$$\text{Substitute: } = \sum x_i y_i - \bar{y}(n\bar{x}) - \bar{x}(n\bar{y}) + n\bar{x} \bar{y} = \sum x_i y_i - n\bar{x} \bar{y} - n\bar{x} \bar{y} + n\bar{x} \bar{y} = \sum x_i y_i - n\bar{x} \bar{y}.$$

$$\boxed{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x} \bar{y}}$$

Prove:  $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = b_1^2 s_x^2 (n-1) = b_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 = b_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

In simple linear regression:  $b_0 = \bar{y} - b_1 \bar{x}$

Substitute:  $\hat{y}_i = b_0 + b_1 x_i = \bar{y} - b_1 \bar{x} + b_1 x_i$

Organize:  $\hat{y}_i - \bar{y} = b_1 (x_i - \bar{x})$

$(\hat{y}_i - \bar{y})^2 = b_1^2 (x_i - \bar{x})^2$

Summing:  $\sum (\hat{y}_i - \bar{y})^2 = b_1^2 \sum (x_i - \bar{x})^2$

$$\boxed{\sum (\hat{y}_i - \bar{y})^2 = b_1^2 \sum (x_i - \bar{x})^2}$$

Sample Variance of  $x$ :  $s_x^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

Multiply both sides by  $n-1$ :  $\sum (x_i - \bar{x})^2 = s_x^2 (n-1)$

Substitute:  $\sum (\hat{y}_i - \bar{y})^2 = b_1^2 s_x^2 (n-1)$

$$\boxed{\sum (\hat{y}_i - \bar{y})^2 = b_1^2 s_x^2 (n-1)}$$

Least-squares slope estimator:  $b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$

Multiply both sides by  $\sum (x_i - \bar{x})^2$ :  $b_1 \sum (x_i - \bar{x})^2 = \sum (x_i - \bar{x})(y_i - \bar{y})$

Multiply both sides by  $b_1$ :  $b_1^2 \sum (x_i - \bar{x})^2 = b_1 \sum (x_i - \bar{x})(y_i - \bar{y})$

Recall:  $\sum (\hat{y}_i - \bar{y})^2 = b_1^2 \sum (x_i - \bar{x})^2$

$$\boxed{\sum (\hat{y}_i - \bar{y})^2 = b_1 \sum (x_i - \bar{x})(y_i - \bar{y})}$$